



Predictive Modeling Checklist

Provide the following information when **making a filing to introduce a new or modify an existing** predictive model, to include but not limited to: Generalized Linear Model (GLM), GBM, Decision Trees and Random Forest.

If filing a usage based insurance, telematics or similar type of behavior model, please refer to the Telematics Checklist found on SERFF

Please send any questions regarding this checklist to: propcas@difi.az.gov- include Predictive Modeling Checklist in the subject line.

NOTE: File the predictive model under the applicable type of insurance (TOI) and filing type Predictive Model. Any associated rates, rating factors or changes to base rates associated with the predictive model are to be filed separately as Rate, or Rate/Rule filing types.

In a descriptive written narrative, provide the following information. Provide requested charts and graphs and Excel workbooks where applicable.

1. MODEL DEVELOPMENT

- a. What is the Predictive Model intended to model? (e.g. Frequency and Severity, Loss Ratios, Pure Premium)
- b. If not purchased from an Rate Service Organization (RSO) or an Advisory Organization, provide the modeler's qualifications and credentials, include the entire modeling team as well as the members of company leadership involved. If purchased from an RSO/Advisory, provide a contact or the SERFF filing number that the model was filed under.
- c. Is this model a re-fresh or a re-build of a previously filed model? If so, also provide the SERFF number/s for the prior version/s. In addition to addressing questions in sections 1-4 refer to additional questions in section 5) Revised Model.
- d. If any external models were used in the building of this model, reference those models and the appropriate SERFF filing number.
- e. Address if this predictive model is intended to interact with an existing model, or is a sub-model.
- f. Reference the modeling software used and why it was selected.
- g. Provide a description of the training, testing, and validation data used.
- h. Describe actuarial standards employed, including any specific ASOPs.
- i. Provide an academic reference for the model type used.
- j. Provide a correlation matrix or heat map in Excel for the variables used in the model.
- k. How was the model tested for stability? Was the risk of overfitting data considered and how was it mitigated?
- l. Describe the methods used to validate the goodness of fit such as lift charts,